

Assessment 2 Individual Workbook

Section 1: Personal contributions to the team design portfolio

User story 1 Name:

Stub: Jazz Instrument Matching Game

User story: As a kindergarten student, I want to match the trumpet to Miles Davis so that I can unlock his cool jazz sound.

Acceptance criteria:

1. The screen displays a cartoon Miles Davis wearing cool sunglasses next to a trumpet.
2. Dragging the trumpet to Miles Davis triggers a sparkle effect and a bright trumpet tune.
3. Incorrect instruments bounce back with a friendly "uh-oh" chime.

Story size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/design/9Jh5qEakj9ugq4MLgMsgnV/Untitled?node-id=4-9&t=MfWI30dGyWnrC4H9-1>

URL to the game:

<https://www.kopo.com/g/game.php?gno=14380&qid=8676&lid=5385&sid=2973>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	Kindergarten student	Learn how to view jazz musicians.	Through this game, I learned about the fascinating stories of jazz musicians.
2	Kindergarten student	this game lets you find out who the most famous jazz drummers are.	I know who the most famous jazz drummers are.
3	Kindergarten student	this game, you can learn about the most famous jazz drummers.	Through this game, I learned about the most famous jazz drummers.
4	Kindergarten	this game explores the	I learned the stories of jazz drummers

	student	story of a jazz drummer.	through this game.
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Feedback

1. We are kindergarten games. Check if Kindergarten students can play
2. Because you are making games for KOPO, you need to show 4 answer images, 4 hint images and 4 drop boxes, buttons, score wheels for KOPO game
3. Make images for child friendly

User story 2 Name:

Stub:Jazz Instrument Matching Game

User story: As a kindergarten student, I want to match the playful piano keys to Thelonious Monk so that I can see him do a fun, rhythmic jazz dance.

Acceptance criteria:

1. The screen displays cartoon Thelonious Monk in a hat next to a piano.
2. Dragging the piano to Monk triggers a fun dancing animation.
3. A friendly voiceover says: "Great job! Thelonious Monk plays the piano!"

Story size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/design/9Jh5qEakj9ugq4MLgMsgnV/Untitled?node-id=4-9&t=MfWI30dGyWnrC4H9-1> _____

URL to the game:

<https://www.kopo.com/g/game.php?gno=14380&qid=8676&lid=5385&sid=2973>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	Kindergarten student	Fun rote counting matching game level	Learn Jass musicians' knowledge while having fun.
2	Kindergarten student	this game introduces some of the most famous jazz pianists.	I learned the stories of the most famous jazz pianists.
3	Kindergarten student	A game helped me learn about musical instruments	I get to know different musical instruments through this game.

4	Kindergarten student	gain more insight into musicians and knowledge about musical instruments.	I gained relevant knowledge about musicians and musical instruments.
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Feedback

1. We are kindergarten games. Check if Kindergarten students can play
2. Because you are making games for KOPO, you need to show 4 answer images, 4 hint images and 4 drop boxes, buttons, score wheels for KOPO game
3. Make images for child friendly

User story 3 Name:

Stub: Coltrane Sax Match

User story: As a kindergarten student, I want to match the large golden saxophone to John Coltrane so that I can easily recognize its shape and hear a smooth sing.

Acceptance Criteria:

1. The screen displays a large, easily recognizable golden saxophone illustration alongside a friendly portrait of John Coltrane.
2. Successfully dragging and dropping the golden saxophone onto John Coltrane plays a smooth 5-second saxophone melody.

Story size: L

URL to prototype (Figma prototype design):

<https://www.figma.com/design/9Jh5qEAKj9ugq4MLgMsgnV/Untitled?node-id=4-9&t=MfWI30dGyWnrC4H9-1> ____

URL to the game:

<https://www.kopo.com/g/game.php?gno=14380&qid=8676&lid=5385&sid=2973>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	Kindergarten student	get to know jazz figures	I can recognize his signature cool jazz sound while enjoying a fun visual challenge.
2	Kindergarten student	a saxophone drag-and-drop matching game	I can learn to distinguish the unique tones of different jazz instruments.

3	Kindergarten student	a matching game that lets you learn about musicians	I can learn more about musicians.
4	Kindergarten student	this game introduces the most famous jazz saxophonists.	I learned about the most famous jazz saxophonists by playing this game.

Feedback

1. Add animations if can
2. Add outlines/strokes to the answer and hints images
3. Improve questions for children

Section 2: Rationale behind key design decisions you were involved in

2.1 Project Goals and Initial Concept

Modern software engineering paradigms have increasingly shifted away from rigid, plan-based specifications toward human-centered, iterative methodologies that prioritize user adaptability and evolving requirements (Sommerville, 2016). Within this framework, the design, conceptualization, and user experience (UX) architecture were established for an interactive educational game designed to match jazz musicians with their respective instruments, targeting early childhood learners. The primary product goal centers on introducing rich cultural music history through gamified learning, effectively reframing abstract historical narratives into accessible, engaging digital interactions (Brown, 2008). Developed as a form of made-to-order software tailored to early childhood educational standards, the system integrates tailored visual and functional components to lower cognitive load while maximizing learner engagement. By embedding agile development practices and human-centered design principles throughout the initial conceptualization phase, the project aligns technical interaction mechanics with foundational pedagogical objectives (Norman, 2013).

2.2 Developing the Game Requirements

To establish a rigorous foundation for the software's requirements engineering process, extensive scholarly research was conducted across twelve iconic jazz figures to guide authentic domain asset elicitation. To accommodate early childhood cognitive capabilities, complex historical photographs were systematically reconceived into vibrant cartoon illustrations, translating abstract cultural heritage into age-appropriate visual representations (Norman, 2013). Parallel to visual asset transformation, technical spikes were executed on the KOPO

platform to evaluate spatial boundaries, architectural constraints, and underlying interaction mechanics, thereby determining both functional and non-functional requirements (Sommerville, 2016). User research inputs were gathered through empathetic, open-ended interview tasks designed to observe children's reactions to musical feedback, error mechanics, and character recognition (Brown, 2008). By synthesizing historical domain research, platform-specific technical experimentation, and direct child-centered empathy exercises, the development team established a comprehensive, user-informed specification set tailored for gamified early childhood education.

2.3 Prototyping and Iterative Improvement

To translate conceptual frameworks into functional mechanics, functional and aesthetic components were synthesized within Figma into clickable interactive prototypes spanning low-fidelity (Lo-fi), medium-fidelity (Mid-fi), and high-fidelity (Hi-fi) stages to establish an intuitive information architecture (IA) (Sommerville, 2016). Evaluated through task testing within a structured 5-Act Interview framework, user observations yielded critical empirical feedback regarding childhood interaction patterns (Brown, 2008). Initial testing revealed hesitations during matching tasks, such as associating piano keys with Thelonious Monk, which were resolved through engaging character animations. Furthermore, non-punitive "bounce-back" animations combined with auditory "uh-oh" chimes successfully mitigated user frustration during incorrect matching attempts, preserving confidence. Conversely, immediate multimodal feedback—such as triggering trumpet melodies upon selecting Miles Davis—provided effective positive reinforcement, driving iterative refinement of the system's interactive feedback loop (Norman, 2013).

2.4 Final Design Decisions and Trade-offs

To establish optimized user interactions, localized, child-friendly narrative backgrounds and thematic visual environments were systematically curated alongside cheerful background music, leveraging Gestalt principles of similarity and proximity to strengthen information architecture and visual organization (Koffka, 1935; Norman, 2013). A key architectural trade-off analysis resulted in refactoring complex archival photographs into simplified cartoon visual styles, effectively minimizing cognitive load while optimizing usability and acceptability for early childhood learners (Sommerville, 2016). This stylistic simplification was strategically balanced with immediate, high-contrast visual and auditory feedback mechanisms—such as dynamic character animations and distinct melody cues—to sustain intrinsic learner engagement and maintain high confidence during task execution. Furthermore, these refined visual assets, multimodal interaction mechanics, and localized narrative components were rigorously validated against the project's explicit Definition of Done (Schwaber & Sutherland, 2020). This systematic verification ensured that all functional, aesthetic, and pedagogical requirements satisfied rigorous software quality standards prior to final deployment.

Section 3: Reflects on feedback, iteration, constraints, and design trade-offs

3.1 Feedback and Iteration

In accordance with agile empiricism—emphasizing transparency, inspection, and adaptation—stakeholder feedback elicited during the Sprint Review provided critical insights into overall software usability and user acceptability (Schwaber & Sutherland, 2020). Serving as a primary domain stakeholder, the course instructor observed that employing realistic historical photographs imposed an excessively high cognitive load on six-year-old learners, thereby prompting a structured iterative refinement cycle. To optimize learner engagement and reduce perceptual friction, the team executed visual refactoring by transforming dense archival portraits into vibrant, accessible cartoon illustrations (Sommerville, 2016). In tandem, complex narrative structures were reframed into simplified story vignettes to align with early childhood literacy levels, while upbeat auditory cues were integrated to deliver positive reinforcement. By systematically embedding empirical stakeholder evaluations into continuous iterative development, the engineering team successfully harmonized core pedagogical objectives with early childhood usability requirements (Brown, 2008).

3.2 Evaluation of Outcomes

To systematically evaluate design outcomes, the software architecture was conceptualized using the Design Thinking double-diamond model of divergent and convergent thinking, carefully tailored to accommodate the developmental requirements of the target six-year-old user persona (Brown, 2008; Piaget, 1962). The underlying information architecture (IA) operationalized fundamental Gestalt principles to optimize visual cognition and spatial interaction: visual proximity was established by positioning musician avatars adjacent to their corresponding instrument drop-zones to minimize cognitive load, while visual similarity was preserved through uniform cartoon aesthetics across all twelve character cards to clearly signal standardized interactive behaviors (Koffka, 1935; Norman, 2013). Interface layouts were engineered using a hybrid grid and drill-down paradigm optimized for small-screen mobile displays, thereby streamlining user navigation. Furthermore, the development lifecycle progressed through systematic multi-fidelity prototyping stages—advancing from low-fidelity paper wireframes to medium-fidelity clickable layouts, and ultimately to high-fidelity Figma mockups embedded with dynamic color palettes and auditory feedback—to thoroughly validate aesthetic and functional requirements prior to final deployment (Sommerville, 2016).

3.3 Constraints and Design Trade-offs

Addressing system constraints required explicit design trade-offs to carefully balance overarching pedagogical goals with early childhood usability and software acceptability. A

primary architectural compromise prioritized cognitive accessibility and developmental appropriateness over strict historical realism, substituting detailed, archival photographs with stylized, simplified cartoon representations to significantly reduce young learners' perceptual processing demands and intrinsic cognitive load (Norman, 2013; Piaget, 1962). To effectively reconcile broad functional scope requirements—which mandated the inclusion of twelve distinct historical jazz figures—with strict delivery time-boxes, strategic component selection and design reuse methodologies were systematically integrated within Figma (Sommerville, 2016). By leveraging modular UI assets and re-usable graphical templates, the engineering team accelerated asset production and interface assembly without compromising the required historical and cultural diversity of the educational content. Furthermore, technical platform constraints were proactively identified through preliminary technical spikes executed on the specialized KOPO platform (Beck & Fowler, 2000). These exploratory spikes enabled the team to evaluate spatial boundaries, hardware rendering capabilities, and architectural limits early in the development lifecycle, thereby preventing downstream integration bottlenecks and ensuring the delivery of a performant, stable early childhood learning system.

3.4 Lessons Learned and Action Plan

Critical reflection during the Sprint Retrospective enabled the development team to systematically identify and correct workflow anti-patterns, specifically transitioning away from passive, one-way demonstration rituals toward genuine collaborative inspection and transparent stakeholder engagement (Schwaber & Sutherland, 2020). To capture robust evidence of learning throughout the software engineering lifecycle, interactive Figma artifacts and multi-fidelity prototypes were leveraged as empirical proof to justify iterative design modifications, validate usability enhancements, and demonstrate quantifiable uncertainty reduction (Kniberg, 2011). This structured reflective practice directly reinforced continuous improvement mechanisms within the agile process, ensuring that actionable feedback derived from empirical observations was seamlessly triaged and incorporated into subsequent Sprint backlogs. By systematically evaluating retrospective outcomes and artifact quality against a rigorous Definition of Done, the cross-functional team progressively matured its self-managing capabilities and internal governance structures (Sommerville, 2016). Ultimately, this agile feedback loop guaranteed that technical software engineering decisions, interface refinements, and interaction mechanics remained deeply aligned with empirical stakeholder insights and early childhood pedagogical requirements throughout all subsequent development cycles.

References

1. Beck, K., & Fowler, M. (2000). Planning extreme programming. Addison-Wesley Professional. <https://www.oreilly.com/library/view/planning-extreme-programming/0201710919/>

2. Beck, K., & Fowler, M. (2000). Planning extreme programming. Addison-Wesley Professional. <https://www.oreilly.com/library/view/planning-extreme-programming/0201710919/>
3. Brown, T. (2008). Design thinking. Harvard Business Review, 86(6), 84–92. <https://hbr.org/2008/06/design-thinking>
4. Brown, T. (2008). Design thinking. Harvard Business Review, 86(6), 84–92. <https://hbr.org/2008/06/design-thinking>
5. Garner, J. K., & Alley, M. (2013). How the design of presentation slides affects audience comprehension: A case study. International Journal of Engineering Education, 29(6), 1564–1579. <https://www.ijee.ie/articles/10.1002/jee.20030>
6. Kniberg, H. (2011). Lean from the trenches: Managing large-scale projects with Kanban. Pragmatic Bookshelf. <https://pragprog.com/titles/hblean/lean-from-the-trenches/>
7. Koffka, K. (1935). Principles of Gestalt psychology. Routledge & Kegan Paul. <https://www.routledge.com/Principles-of-Gestalt-Psychology/Koffka/p/book/9780415209625>
8. Norman, D. (2013). The design of everyday things (Revised and expanded ed.). Basic Books. <https://www.basicbooks.com/titles/don-norman/the-design-of-everyday-things/9780465050659/>
9. Norman, D. (2013). The design of everyday things (Revised and expanded ed.). Basic Books. <https://www.basicbooks.com/titles/don-norman/the-design-of-everyday-things/9780465050659/>
10. Piaget, J. (1962). Play, dreams and imitation in childhood. W. W. Norton & Company. <https://wnorton.com/books/9780393001716>
11. Schwaber, K., & Sutherland, J. (2020). The Scrum Guide: The definitive guide to Scrum: The rules of the game. Scrum.org. <https://scrumguides.org/scrum-guide.html>
12. Schwaber, K., & Sutherland, J. (2020). The Scrum Guide: The definitive guide to Scrum: The rules of the game. Scrum.org. <https://scrumguides.org/scrum-guide.html>

13. Sommerville, I. (2016). Software engineering (10th ed.). Pearson.
<https://www.pearson.com/en-us/subject-catalog/p/software-engineering/P200000003303>
14. Sommerville, I. (2016). Software engineering (10th ed.). Pearson.
<https://www.pearson.com/en-us/subject-catalog/p/software-engineering/P200000003303>
15. Tuckman, B. W. (1965). Developmental sequence in small groups. Psychological Bulletin, 63(6), 384–399. <https://doi.org/10.1037/h0022100>

Assessment 2: Rubric — Individual Design Workbook (15%)

Criterion	HD (85–100%)	D (75–84%)	C (65–74%)	P (50–64%)	F (0–49%)
Identification of individual contribution 5%	Clearly and convincingly articulates specific individual contributions to the team design portfolio, demonstrating strong ownership and understanding of how those contributions shaped the overall design.	Clearly identifies individual contributions with minor gaps in specificity or clarity.	Identifies individual contributions at a general level, with limited detail.	Contributions are vaguely described or difficult to distinguish from overall teamwork.	Individual contribution is unclear, implausible, or not demonstrated.
Design rationale and reflective reasoning 5%	Provides insightful and well-justified explanations of design decisions, demonstrating reflective reasoning, awareness of alternatives, and consideration of constraints or trade-offs.	Provides clear justification of design decisions with some reflective insight and awareness of alternatives.	Provides basic explanation of design decisions with limited reflection or justification.	Provides minimal or largely descriptive explanation of decisions, with little reflection.	Provides little or no meaningful explanation or reflection on design decisions.
Professional communication and academic clarity 5%	Writing is clear, well structured, and professional; design and software engineering terminology is used appropriately; referencing is accurate where required.	Writing is clear and professional with minor issues; terminology is mostly appropriate; referencing is mostly accurate.	Writing is generally clear but uneven or loosely structured; terminology or referencing may be inconsistent.	Writing lacks clarity or structure; terminology is used inaccurately; referencing is weak or inconsistent.	Writing is unclear or unprofessional; terminology is inappropriate; referencing is absent or incorrect.

